



```
In [ ]: import nltk
```

```
nltk.download('brown')  
nltk.download('universal_tagset')
```

```
[nltk_data] Downloading package brown to /root/nltk_data...  
[nltk_data]   Unzipping corpora/brown.zip.  
[nltk_data] Downloading package universal_tagset to /root/nltk_data...  
[nltk_data]   Unzipping taggers/universal_tagset.zip.
```

```
Out[ ]: True
```

```
In [ ]: from nltk.corpus import brown
```

```
sentences = brown.sents()  
print("Total sentences:", len(sentences))
```

```
Total sentences: 57340
```

```
In [ ]: tokens = [word.lower() for sent in sentences for word in sent]
```

```
print("Total tokens:", len(tokens))  
print("Sample tokens:", tokens[:20])
```

```
Total tokens: 1161192
```

```
Sample tokens: ['the', 'fulton', 'county', 'grand', 'jury', 'said', 'friday',  
'an', 'investigation', 'of', 'atlanta's', 'recent', 'primary', 'election', 'pro  
duced', '``', 'no', 'evidence', '``', 'that']
```

```
In [ ]: tagged_sentences = brown.tagged_sents(tagset='universal')
```

```
print("Sample tagged sentence:")  
print(tagged_sentences[0])
```

```
Sample tagged sentence:
```

```
[('The', 'DET'), ('Fulton', 'NOUN'), ('County', 'NOUN'), ('Grand', 'ADJ'), ('Ju  
ry', 'NOUN'), ('said', 'VERB'), ('Friday', 'NOUN'), ('an', 'DET'), ('investigat  
ion', 'NOUN'), ('of', 'ADP'), ('Atlanta's', 'NOUN'), ('recent', 'ADJ'), ('prima  
ry', 'NOUN'), ('election', 'NOUN'), ('produced', 'VERB'), ('``', '.'), ('no',  
'DET'), ('evidence', 'NOUN'), ('``', '.'), ('that', 'ADP'), ('any', 'DET'), ('i  
rregularities', 'NOUN'), ('took', 'VERB'), ('place', 'NOUN'), ('.', '.')] 
```

```
In [ ]: from nltk.tag.hmm import HiddenMarkovModelTrainer
```

```
trainer = HiddenMarkovModelTrainer()  
hmm_model = trainer.train_supervised(tagged_sentences)  
  
print("HMM Model trained successfully!")
```

```
HMM Model trained successfully!
```

```
In [ ]: transition_prob = hmm_model._transitions
```

```
print("Transition Probability Example:")  
for prev_tag in list(transition_prob.keys())[:3]:  
    print(prev_tag, "→", transition_prob[prev_tag])
```

Transition Probability Example:

DET → <MLEProbDist based on 137001 samples>

NOUN → <MLEProbDist based on 274644 samples>

ADJ → <MLEProbDist based on 83690 samples>

```
In [ ]: emission_prob = hmm_model._outputs

print("Emission Probability Example:")
for tag in list(emission_prob.keys())[:3]:
    print(tag, "→", emission_prob[tag])
```

Emission Probability Example:

DET → <MLEProbDist based on 137019 samples>

NOUN → <MLEProbDist based on 275558 samples>

ADJ → <MLEProbDist based on 83721 samples>

```
In [ ]: def predict_pos(sentence):
        tokens = [word.lower() for word in sentence.split()]
        predicted_tags = hmm_model.tag(tokens)
        return predicted_tags
```

```
In [ ]: test_sentence = "The quick brown fox jumps over the lazy dog"

prediction = predict_pos(test_sentence)

print("Sentence:", test_sentence)
print("Predicted POS Tags:")
for word, tag in prediction:
    print(f"{word} → {tag}")
```

/usr/local/lib/python3.12/dist-packages/nltk/tag/hmm.py:335: RuntimeWarning: overflow encountered in cast

```
0[i, k] = self._output_logprob(si, self._symbols[k])
```

Sentence: The quick brown fox jumps over the lazy dog

Predicted POS Tags:

the → DET

quick → ADJ

brown → ADJ

fox → NOUN

jumps → VERB

over → ADP

the → DET

lazy → ADJ

dog → NOUN